

Binary Detection

June 16, 2026

1 Class Assignments

1.1 Assignment 1: MAP Threshold and Error Probability

Objective. Verify the analytical maximum a posteriori (MAP) detector for binary signal detection in additive white Gaussian noise (AWGN).

Problem statement. Consider the binary hypothesis-testing problem

$$H_0 : Y = a_0 + \eta, \quad (1)$$

$$H_1 : Y = a_1 + \eta, \quad (2)$$

where

$$\eta \sim \mathcal{N}(0, \sigma_\eta^2), \quad (3)$$

and the prior probabilities are

$$P(H_0) = \pi_0, \quad P(H_1) = \pi_1, \quad \pi_0 + \pi_1 = 1. \quad (4)$$

Tasks.

1. Derive the MAP decision rule for deciding between H_0 and H_1 .
2. Show that the MAP detector can be written as a threshold test:

$$Y \underset{H_0}{\overset{H_1}{\geq}} \gamma_{\text{MAP}}. \quad (5)$$

3. Derive the analytical expression for the MAP threshold γ_{MAP} for the equal-variance Gaussian case.
4. Implement the MAP detector in Python.
5. Sweep the decision threshold γ over a suitable interval and compute the probability of error

$$P_e(\gamma) = \pi_0 P_{\text{FA}}(\gamma) + \pi_1 P_{\text{M}}(\gamma), \quad (6)$$

where P_{FA} is the false-alarm probability and P_{M} is the miss probability.

6. Compare the numerically optimal threshold with the analytical MAP threshold.

Required deliverables.

1. Derivation of the MAP likelihood-ratio test.
2. Analytical expression for γ_{MAP} .
3. Plot of $P_e(\gamma)$ versus γ .

4. Table reporting:

Quantity	Value
γ_{MAP}	
Numerically optimal threshold	
P_{FA}	
P_{M}	
P_e	

5. Brief discussion of agreement or disagreement between the analytical and numerical results.